

KAUHAHA, Finland

WMO: 029130

Lat: 63.136N

Lon: 23.053E

Elev: 46

StdP: 100.77

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-26.0	-22.8	-28.4	0.3	-25.9	-25.0	0.4	-22.7	10.4	-1.4	9.2	-0.8	1.1	50	0.653

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	26.6	18.2	24.7	17.3	22.9	16.3	19.7	24.1	18.5	22.7	17.4	21.4	3.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.1	21.4	16.9	12.1	20.4	15.7	11.2	19.1	56.4	24.0	52.3	22.6	49.0	21.3	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	8.0	7.1	DB	-28.6	29.0	4.9	2.1	-32.1	30.6	-35.0	31.8	-37.7	33.0	-41.2	34.6	(4)
(5)			WB	-28.6	21.2	4.8	1.6	-32.1	22.4	-34.9	23.3	-37.6	24.2	-41.0	25.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.0	-7.1	-7.3	-3.3	3.3	8.7	13.5	16.4	14.5	9.8	4.1	-0.5	-4.3	(6)
(7)		DBStd	9.52	7.00	7.22	5.27	3.73	4.23	3.38	3.08	3.16	3.55	4.07	4.88	6.52	(7)
(8)		HDD10.0	2708	530	486	413	204	75	7	1	3	47	187	315	443	(8)
(9)		HDD18.3	5243	788	719	671	452	299	149	77	123	256	442	565	701	(9)
(10)		CDD10.0	533	0	0	0	2	36	112	197	143	41	2	0	0	(10)
(11)		CDD18.3	27	0	0	0	0	1	4	16	6	0	0	0	0	(11)
(12)		CDH23.3	318	0	0	0	0	22	56	174	66	0	0	0	0	(12)
(13)		CDH26.7	51	0	0	0	0	2	9	33	6	0	0	0	0	(13)
(14)	Wind	WSAvg	3.2	3.2	3.2	3.4	3.3	3.5	3.3	2.7	2.7	3.0	3.2	3.4	3.4	(14)
(15)	Precipitation	PrecAvg	560	36	27	27	28	42	61	75	66	52	54	47	42	(15)
(16)		PrecMax	691	58	49	47	70	79	119	122	122	138	116	90	73	(16)
(17)		PrecMin	432	13	2	4	6	12	20	23	21	18	16	11	18	(17)
(18)		PrecStd	65	13	14	11	15	20	25	29	29	33	26	20	16	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.5	5.6	9.8	18.5	26.0	28.1	29.9	27.3	21.3	14.2	10.0	7.1	(19)
(20)			MCWB	3.6	3.5	5.3	10.2	15.2	18.2	20.2	18.8	15.7	12.0	9.1	6.7	(20)
(21)		2%	DB	3.2	3.7	6.8	15.1	22.9	24.8	27.3	25.2	19.2	12.3	8.2	5.4	(21)
(22)			MCWB	2.4	2.4	3.6	8.8	14.1	16.5	18.8	17.8	14.7	10.8	7.5	4.9	(22)
(23)		5%	DB	2.2	2.2	4.9	12.3	20.0	22.6	25.2	23.1	17.4	11.1	7.0	4.1	(23)
(24)			MCWB	1.6	1.6	2.4	7.1	12.6	15.1	18.0	17.0	13.6	9.9	6.5	3.4	(24)
(25)		10%	DB	1.2	1.2	3.2	10.1	17.0	20.8	23.2	21.2	15.9	10.0	5.8	3.0	(25)
(26)			MCWB	0.8	0.7	1.4	6.1	10.8	14.4	17.1	16.1	13.0	8.9	5.3	2.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.9	3.7	5.9	11.6	16.7	19.7	21.8	21.0	16.7	12.5	9.2	6.6	(27)
(28)			MCDWB	4.1	4.9	9.1	16.4	23.1	24.3	27.1	25.1	19.8	13.7	9.9	6.8	(28)
(29)		2%	WB	2.6	2.5	3.9	9.5	15.1	17.9	20.3	19.1	15.4	11.3	7.5	5.0	(29)
(30)			MCDWB	2.9	3.3	6.1	14.1	21.0	23.2	25.0	22.8	18.1	12.2	8.0	5.2	(30)
(31)		5%	WB	1.6	1.7	2.7	7.8	13.2	16.3	19.0	17.8	14.4	10.2	6.4	3.5	(31)
(32)			MCDWB	2.0	2.2	4.2	11.5	19.0	20.6	23.3	21.5	16.6	11.1	6.8	4.0	(32)
(33)		10%	WB	0.8	0.8	1.6	6.2	11.4	15.1	18.0	16.7	13.5	8.9	5.2	2.4	(33)
(34)			MCDWB	1.2	1.1	2.9	9.7	16.2	19.4	22.1	20.3	15.5	9.7	5.6	2.8	(34)
(35)	Mean Daily Temperature Range		MDBR	6.6	7.3	8.7	9.2	10.9	10.2	10.4	10.5	9.2	6.0	4.5	5.7	(35)
(36)		5% DB	MCDBR	5.1	5.7	9.1	13.8	16.0	14.0	14.0	13.9	11.4	6.8	4.9	4.5	(36)
(37)			MCWBR	4.9	4.9	6.2	8.3	8.2	7.0	7.0	7.4	7.2	5.4	4.7	4.4	(37)
(38)		5% WB	MCDBR	5.1	5.3	7.9	12.4	14.3	11.7	12.1	11.7	9.6	6.1	4.8	4.5	(38)
(39)			MCWBR	5.1	4.7	5.7	7.9	7.8	6.5	6.5	6.8	6.4	5.1	4.6	4.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.232	0.290	0.316	0.356	0.352	0.351	0.373	0.359	0.333	0.315	0.379	0.353	(40)	
(41)		taud	2.041	2.169	2.253	2.333	2.431	2.456	2.416	2.451	2.509	2.385	2.688	2.593	(41)	
(42)		Ebn at Noon	390	636	778	824	864	870	837	818	771	619	361	217	(42)	
(43)		Edh at Noon	31	63	86	100	99	98	100	88	69	52	28	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.16	0.73	2.23	3.72	5.02	5.47	5.07	3.89	2.27	0.88	0.22	0.07	(44)	
(45)		RadStd	0.02	0.15	0.23	0.38	0.54	0.48	0.49	0.40	0.25	0.17	0.04	0.01	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.70	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page